$f(x) = |x - 1|$ causes the graph of the function $f(x) = |x|$ shifts to the right for one unit.

$f(x) = |x + 1|$ causes the graph of the function $f(x) = |x|$ shifts to the left for one unit. (See Figure 1.51.)

Illustration 1.9 Consider the functions:

$$\text{(i) } f(x) = |x| \text{ (ii) } f(x) = |x| - 1 \text{ (iii) } f(x) = |x| + 1$$

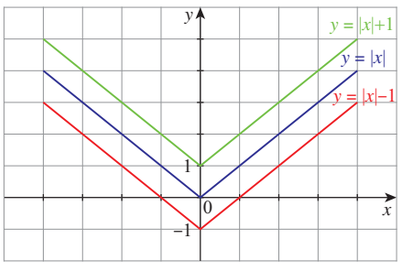


Figure 1.52

$f(x) = |x| - 1$ causes the graph of the function $f(x) = |x|$ shifts to the downward for one unit.

$f(x) = |x| + 1$ causes the graph of the function $f(x) = |x|$ shifts to the upward for one unit. (See Figure 1.52.)

Dilation

Dilation is also a transformation which causes the curve stretches (expands) or compresses (contracts). Multiplying a function by a positive constant vertically stretches or compresses its graph; that is, the graph moves away from x-axis or towards x-axis.

If the positive constant is greater than one, the graph moves away from the x-axis. If the positive constant is less than one, the graph moves towards the x-axis.

Illustration 1.10 Consider the functions:

$$\text{(i) } f(x) = x^2 \text{ (ii) } f(x) = \tfrac{1}{2}x^2 \text{ (iii) } f(x) = 2x^2$$

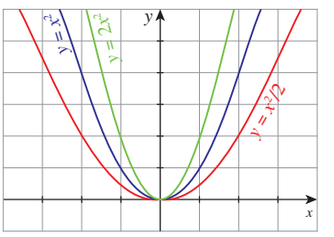


Figure 1.53

$f(x) = \frac{1}{2}x^2$ causes the graph of the function $f(x) = x^2$ stretches towards the x-axis since the multiplying factor is $\frac{1}{2}$ which is less than one.

$f(x) = 2x^2$ causes the graph of the function $f(x) = x^2$ compresses towards the $y-$ axis that is, moves away from the $x-$axis since the multiplying factor is 2 which is greater than one. (See Figure 1.53.)